



A strategy for quantifying microplastic particles in membrane filtration processes using flow cytometry

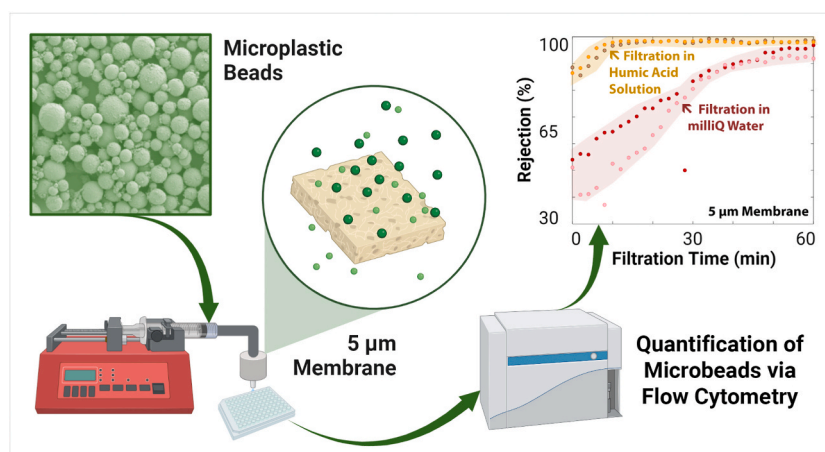
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HIGHLIGHTS

- Fluorescent microplastic particle concentrations were measured via flow cytometry.
- The presence of autofluorescent humic acid did not hinder microplastic detection.
- Microfiltration membranes demonstrated a high rejection of microplastic particles.
- Membrane fouling by humic acid mitigated microplastic particle transmission.
- Smaller particle size distributions in membrane permeate vs. feed samples were seen.

GRAPHICAL ABSTRACT



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ABSTRACT

Microplastic (MP) pollution is ubiquitous in the aquatic environment, with significant quantities of MPs originating from municipal wastewater treatment plants. Efforts to evaluate and implement MP removal processes are underway, with membrane technologies often recommended as an “ideal” solution. A key challenge in evaluating these technologies involves efficiently quantifying MP concentrations in samples. Here, flow cytometry (FC) is demonstrated as an effective technique to obtain concentration measurements of plastic microbeads (MBs; 1–5 µm) suspended in water with/without added humic acid. Regardless of solution conditions, MB concentrations were easily quantified via FC. Subsequently, two microfiltration membranes were challenged to these suspensions. As measured via FC, the 0.45 µm membrane demonstrated effective MB rejection (>99%) whereas the 5 µm membrane exhibited a broad range of rejections (40% to >95%) depending on solution conditions and filtration time. Finally, a model was formulated utilizing FC forward light scattering intensity measurements to estimate MB sizes in samples. Using the model, a 33% reduction in median MB size, on average, was noted across the 5 µm membrane when filtering MBs suspended in humic acid solution, affirming a preferential permeation of

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smaller particles. Overall, this study advances MP quantification techniques towards validating removal processes.

1. Introduction

Microplastics (MP) are ubiquitous in the aquatic environment: they have been detected globally, from the most densely populated urban locations to the most remote arctic waters (Kanhai et al., 2020). These particles are often defined as being less than 5 mm in size, however, they can be found in the micrometer size range. Nanoplastics (NPs) are even smaller (e.g. $<1\ \mu\text{m}$) and can originate from the degradation of larger plastic pieces (Koutnik et al., 2021; Enfrin et al., 2021). While the physical (e.g. size, shape, etc.) and chemical (e.g. polymer identity) properties of MPs vary widely, these particles create a persistent biological hazard in marine ecosystems. Numerous researchers have shown the wide-reaching negative ecological consequences associated with MP pollution with potential implications towards human health, commercial interests, and the economy (Avio et al., 2017; Van Cauwenberghe and Janssen, 2014). Smaller-sized MPs may pose particularly concerning ecological consequences due to their high specific surface areas, ability to act as both sources/sinks for contaminants, and their elevated risk of ingestion by aquatic organisms such as fish, plankton, and filter-feeders (Enfrin et al., 2021; Bakir et al., 2014; Chae and An, 2017; da Costa et al., 2016; Hahladakis et al., 2018; Qiao et al., 2019; Liu et al., 2021; Alimi et al., 2018).

A significant point source of these aquatic MPs comes from effluents discharged by wastewater treatment plants (WWTPs) (Sun et al., 2019). To prevent this efflux, research is underway to validate and implement technologies which are effective towards the removal of MPs from effluents. The efficacy of processes such as coagulation/flocculation, oxidation, media filtration, and membrane technologies for MP removal has been recently studied (Lapointe et al., 2020; Wang et al., 2021; Michielssen et al., 2016; Talvitie et al., 2017a; Enfrin et al., 2020). Membrane technologies are often touted as the “ideal solution” for retaining MPs due to their small pores and because they are already validated in WWTPs for the removal of organic matter and nutrients. Anecdotal and deliberate investigations have yielded results which suggest membranes are effective—but not perfect—at MP capture (Michielssen et al., 2016; Foglia et al., 2019; Olmos et al., 2019; Simon et al., 2018; Lv et al., 2019; Talvitie et al., 2017b). Yet, the MP removal performance of these technologies at both laboratory and process scales still requires careful evaluation. Such assessments heavily rely on accurately measuring MP concentrations in water and wastewater samples, but such research is constrained by the techniques used for MP analysis. Pursuant to the theory that membranes can act as “absolute barriers” to MPs, removal or rejection percentages should be very high, implying that MP concentrations in the water leaving the process (e.g. permeate stream) should be exceedingly low. This means that any tool used to quantify MP concentrations in membrane permeate samples must be able to resolve very low particle concentrations.

Researchers currently employ a variety of techniques to quantify MP concentrations. Traditionally, optical microscopy has been heavily deployed to analyze MPs in samples, yielding useful data on concentrations and properties like shape and size. However, the process is labor-intensive due to rigorous sample preparation requirements (e.g. isolating MPs from suspended/dissolved solids in water samples) and manual examination of the individual particles (Alimi et al., 2018). To accelerate MP quantification in both environmental and laboratory-generated samples, investigators have leveraged the ability of suspended particles to absorb and scatter incident light. Spectrophotometry (Enfrin et al., 2022), turbidity measurement (Lapointe et al., 2020), and nanoparticle tracking analysis (Pramanik et al., 2021) techniques have proven useful in measuring MP concentrations. Enhanced MP identification can be achieved by staining or dyeing the

particles with a fluorescent agent, and then using a fluorescent plate reader to quantify MP concentrations as the emitted fluorescence from a sample can be correlated to the amount of MPs present (Maes et al., 2017; Nel et al., 2021; Erni-Cassola et al., 2017; LaRue et al., 2024). A similar quantification method was developed involving the surface labelling of NPs with palladium and subsequent quantification by inductively coupled plasma-mass spectrometry (ICP-MS) (Pulido-Reyes et al., 2022). While these techniques increase the pace by which MP concentrations can be obtained from samples, they are poorly suited to providing valuable information regarding other MP properties such as size or shape which can normally be obtained via microscopy. Various more complex techniques have also been employed, providing greater degrees of sensitivity and information (e.g. chemical identity), while directly quantifying the presence of MPs. Such techniques include gas or liquid chromatography, mass spectrometry, or combinations thereof (Wang et al., 2021; Dümichen et al., 2015; Fries et al., 2013; Elert et al., 2017; Caputo et al., 2021; Gao et al., 2023). However, these techniques lack the efficiency of previously-mentioned methods as extensive sample preparation is required and may be confounded by non-MP matter present in complex water matrices.

As a promising alternative to these methods, flow cytometry (FC) is well-positioned to be a facile, rapid, and high-resolution tool for MP quantification. As a mature technique, it is heavily deployed in biomedical, bioprocessing, and microbiological applications to count a variety of different cell types (Díaz et al., 2010; Wang et al., 2010). In the environmental sciences, complex samples of water and wastewater are analyzed via FC for the purposes of microbial monitoring (Wang et al., 2010). As described by Díaz et al. (2010), a typical flow cytometer accepts a liquid sample containing particles of interest (e.g. cells), whereby hydrodynamic focusing and a sheath fluid are used to carefully align the particles, single file, within a flow cell. This prepares them for individual interrogation by one or more excitation lasers positioned orthogonal to the flow. When illuminated by a laser, a fluorophore-containing particle may emit light, the intensity of which is measured by the instrument (Díaz et al., 2010). The intensity of light scattered by the particle in the forward ($\sim 0.5\text{--}2^\circ$) and side directions ($\sim 15\text{--}135^\circ$) relative to the incident laser beam can be measured simultaneously. This forward scattering (FSC) intensity is related to particle size, while side scattering (SSC) intensity measurements are sensitive to particle “granularity” or “complexity” (Tzur et al., 2011). These measurements can produce information on the identity, concentration, and properties of the analyzed particles.

In MP-related applications, FC has been leveraged to quantify the ingestion of inherently fluorescent MPs by microorganisms (Woods et al., 2018; Long et al., 2017) and investigate the presence of fluorescent microspheres in biofilm samples (Sgier et al., 2016). Kaile et al. (2020) further showed that non-fluorescent MP polymers could be stained with Nile Red dye ($10\ \mu\text{g mL}^{-1}$ in 10% dimethyl sulfoxide) to promote detection via FC. Furthermore, Wang et al. (2021) demonstrated the utility of FC to detect monodisperse ($3\ \mu\text{m}$) fluorescent MP spheres fed at concentrations ranging from $(1.5\text{--}3) \times 10^5\ \text{MP L}^{-1}$ (Wang et al., 2021). Their results indicated that FC can be reliably used in coagulation, oxidation, and media filtration studies. Apart from the ferrate oxidation process which could yield high MP removal (e.g. $>95\%$) at dosages above $10\ \text{mg Fe L}^{-1}$, these processes tended to yield only modest MP removal in most cases (e.g. $<70\%$ removal via media filtration). In contrast, MP filtration experiments using membranes would be expected to produce very high MP removal and very low MP concentrations in permeate samples (LaRue et al., 2024). Building upon these efforts, there are still significant opportunities to carefully validate and exploit FC in experiments where MP rejection is expected to be high,

and with wastewater-centric solution conditions where phenomena such as synergistic fouling between MPs and natural organic matter (NOM) can be quantified. In an extension of the FC technique, [Hyeon et al. \(2024\)](#) utilized a specialized FlowCam dynamic particle imaging cytometer to investigate factors affecting the rejection of MP fibers via specialized silicon carbide and alumina membranes, both with nominal pore sizes of 0.10 μm . Rejection coefficients ranging from 87.7–98.9% and 90.3–97.7%, respectively, were measured for the two membranes. The FlowCam device concurrently obtained images of the fibers to derive valuable property data such as fiber curl, straightness, length, thickness, and aspect ratio—properties which were subsequently correlated to fiber rejection.

Even in the absence of a specialized FlowCam device, there is also good reason to believe that FC could also be exploited to concurrently provide estimates of MP particle size in addition to particle counts: FSC light intensity increases with increasing particle size ([Díaz et al., 2010](#); [Tzur et al., 2011](#); [Foladori et al., 2008](#); [Julià et al., 2000](#)). In fact, polymer microparticle standards of various sizes are routinely used to calibrate FC instruments for the purposes of cell sizing ([Tzur et al., 2011](#); [Chandler et al., 2011](#)). [Julià et al. \(2000\)](#) developed a procedure to correlate FC FSC intensity data to bacterial size; their correlation takes the form of a second-order function. Other researchers have determined their own correlations relating FSC intensity to particle size for the purposes of measuring bacterial biovolume, estimating the size of plankton, and for sorting cells based on their volumes ([Tzur et al., 2011](#); [Foladori et al., 2008, 2010](#); [García et al., 2018](#); [Jacobi et al., 2018](#)). There is an opportunity to extend this methodology towards rapidly estimating the size of “environmental” MPs.

To this end, our work exploits and validates FC towards quantifying concentrations of 1–5 μm fluorescent spherical plastic microbead (MB) particles in removal experiments using microfiltration (MF) membranes. Experiments were performed in both pure milliQ water and a solution of humic acid (HA), a ubiquitous component in NOM and commonly used as a hydrophobic membrane foulant ([Jacobi et al., 2018](#); [Vakondios et al., 2014](#); [Murphy et al., 2013](#)). A FC instrument and a ubiquitous fluorometric plate were compared head-to-head with respect to their MB quantification performance. FC was further utilized to characterize permeate water samples from filtration experiments using Durapore® MF membranes with 0.45 μm and 5 μm nominal pore sizes. Finally, a

novel model which correlates FC FSC intensity to MB size was developed to provide estimates of particle size in conjunction with particle concentration measurements.

2. Materials & methods

2.1. Chemicals

High-purity milliQ water with a resistivity of 18.2 $\text{M}\Omega\text{ cm}$ was produced by an EMD Millipore milliQ Reference system. Acetone and ethanol (Caledon Laboratories Ltd.) were used as received. Humic acid (HA; Lot = STBK1706) was obtained from Sigma-Aldrich, and stock solutions of this HA were prepared in milliQ water. [Vakondios et al. \(2014\)](#) assert that HA concentrations can vary throughout a WWTP (or between facilities), but a good estimate for the concentration of HA in a WWTP's secondary effluent is $\sim 9\text{ mg L}^{-1}$. Thus, a round value of 10 mg L^{-1} was chosen for HA solutions in this study. These HA solutions were thoroughly ultrasonicated (Branson) before use.

Green fluorescent polymer MBs (FMG series; $\lambda_{\text{ex}}/\lambda_{\text{em}} \approx 414/515\text{ nm}$) were obtained from Cospheric (Santa Barbara, California) as a dry powder. The exact polymer chemistry is not specified, but the manufacturer states that the MB is composed of cross-linked, inert thermoset amino formaldehyde polymer. The diameter of the particles is reported as 1–5 μm . An image of the MBs obtained via field-emission scanning electron microscopy (FESEM) can be found in [Fig. 1A](#) along with a histogram of the MB particle diameter in [Fig. 1B](#). Additional details regarding FESEM and particle sizing can be found in the Supporting Information.

100 mg L^{-1} stock suspensions of MBs were produced in milliQ water or 10 mg L^{-1} HA solution and ultrasonicated for at least 30 min to mitigate any aggregation. Just prior to use in subsequent experiments, these stock suspensions were diluted to their final concentrations using either milliQ water or 10 mg L^{-1} HA solution and were ultrasonicated once more for at least 30 min. No surfactant was used as a dispersant.

2.2. Filtration experiments

EMD Millipore Durapore® polyvinylidene difluoride (PVDF) MF membranes were obtained. Pore sizes of 0.45 μm and 5 μm were chosen

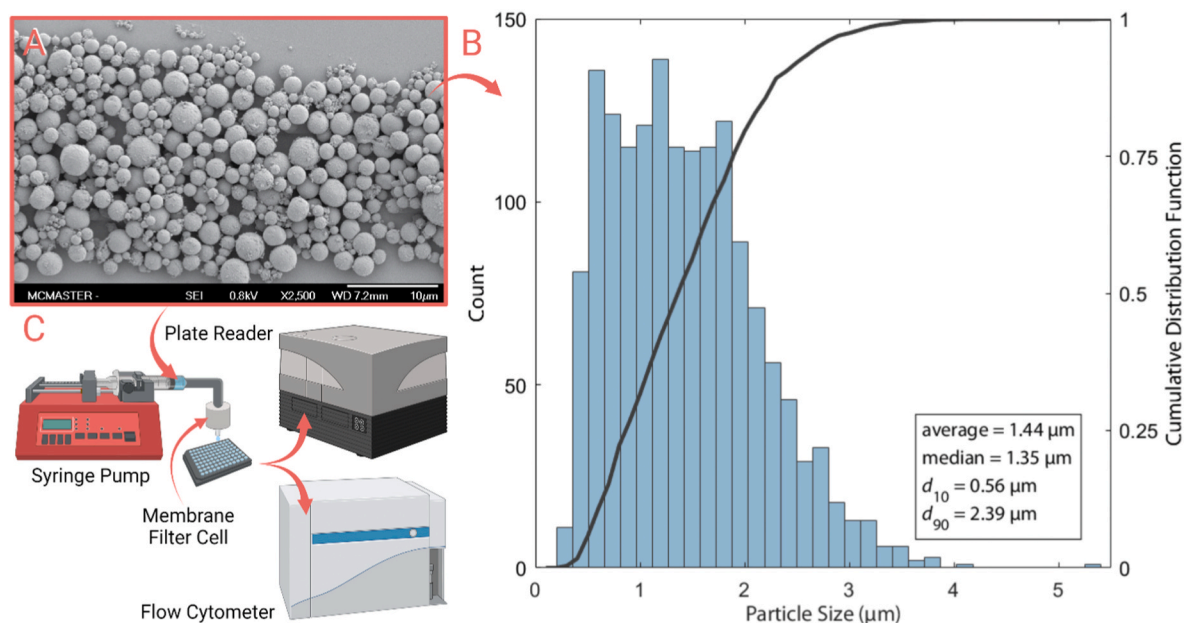


Fig. 1. Characterization and use of the MBs from this study. (A) Sample FESEM image of the MBs. (B) MB particle size distribution and cumulative distribution function ($N = 1,580$). All FESEM images from which these measurements originate are shown in [Figure S1](#). (C) Schematic of the experimental apparatus and post-filtration analysis process.

to be representative of membranes which should strongly reject and poorly reject the MBs used in this study. Pre-wetted membrane discs were subsequently assembled into 13 mm-diameter filter holders (Cole-Parmer) giving an estimated active membrane area of 75 mm². Dead-end filtration experiments with suspensions of MBs (1 mg L⁻¹) in either milliQ water or 10 mg L⁻¹ HA solution were conducted for 60 min at a constant permeate flux of 1000 L m⁻² h⁻¹ (LMH) using a syringe pump (Harvard Apparatus PHD ULTRA). Permeate samples (~300 µL) were collected every 2 min and aliquoted into 96-well microplates for further analysis, as illustrated in Fig. 1C. Information regarding the filtration experimental protocols and a comprehensive characterization of the membranes (e.g. FESEM, porometry) are detailed elsewhere (LaRue et al., 2024).

2.3. Microbead concentration measurements via plate reader

Water samples were analyzed for their MB content using a Tecan Spark 10M plate reader. The instrument gain was optimized at the highest concentration of MBs which was used (1 mg L⁻¹). This optimized gain of “55” was subsequently used in the analysis of all MB-containing samples, with λ_{ex} = 485 nm and λ_{em} = 525 nm (bandwidths = 25 nm and 20 nm, respectively).

2.4. Microbead concentration measurements via flow cytometer

A Beckman Coulter Cytoflex LX flow cytometer was employed to measure the concentrations of MBs in samples. The instrument’s “blue” laser (λ_{ex} = 488 nm) and “FITC” filter (λ_{em} = 525 nm) were selected based on the fluorophore incorporated within the MBs. From the analysis of MB stock suspensions, the quasi-optimal gains corresponding to FSC signal intensity, SSC signal intensity, and the fluorescence (FITC) signal intensity were determined empirically; these values were fixed and used in each subsequent experiment. The gains and other operational parameters are summarized in Table S1 of the Supporting Information. In a given analysis, 50 µL aliquots were drawn by the instrument from a round-bottomed 96-well microplate, allowing for MB counts to be recorded over a known volume. To ensure the validity and reproducibility of the results that were obtained, the instrument was calibrated daily using MB standards according to manufacturer specifications. Furthermore, one well from each microplate was designated as a negative control, containing only milliQ water.

Results recorded via FC were analyzed using FlowJo software (Becton Dickinson). Empirically, gates were developed from FSC signal intensity by peak area (FSC-A; arbitrary units) versus FITC fluorescence signal intensity by peak area (FITC-A; arbitrary units) plots to isolate events corresponding to MBs from events corresponding to background noise or other contaminating particles. Samples of MBs in milliQ water versus milliQ alone, as well as MBs in HA solution versus HA alone were compared to determine the gated region which was expected to contain just events corresponding to the MBs. Coordinates of the polygonal gate bounding these regions are found in Table S1 of the Supporting Information. For the purposes of consistency, these coordinates were used in every FC analysis in this work.

2.5. Microbead size estimation

The methodology developed by Julià et al. (2000) was adapted to provide size (i.e. diameter) estimates of MBs via FC. Define $F_i(i)$ to be the cumulative distribution function (CDF) of the FSC-A intensities i associated with the MBs used in these experiments, as determined via FC, where $0 \leq F_i \leq 1$. Similarly define $F_d(d)$ as the analogous CDF of the particle diameters d of MBs used in these experiments, determined using an *ex situ* method. The premise of the methodology is given that the F_i and F_d model distributions describe the same particle population, then:

$$F_i(i^*) = F_d(d^*) \quad (1)$$

where any individual particle of size d^* produces an FSC-A intensity of i^* , noting that the relationship requires i to increase as d increases which is typically noted in the literature (Erni-Cassola et al., 2017; Hyeon et al., 2024).

The particle size distribution of the MBs was obtained *ex situ* through image analysis of electron micrographs (see §2.2; Fig. 1). Particle sizes $d \in [d_1, d_2, \dots, d_N]$ were recorded ($N = 1,580$) and $F_d(d)$ was subsequently calculated at each value of d . Similarly, a standard 1 mg L⁻¹ sample of MBs in milliQ water was analyzed via FC. The FSC-A intensities $i \in [i_1, i_2, \dots, i_M]$ for the gated events were obtained ($M = 9,429$) from the FlowJo software and $F_i(i)$ was calculated at each value of i . These two CDFs represent the data required to the train the model. A new function $d = \varphi(i)$ was computed which maps the measured i values to their corresponding measured d values using the CDFs from Equation (1):

$$F_i(i) = F_d(\varphi(i)) \quad (2a)$$

$$\varphi(i) = (F_d^{-1} \circ F_i) \quad (2b)$$

The function $d = \varphi(i)$ realizes the transformation of measured FSC-A intensities into estimated particle sizes.

3. Results & discussions

3.1. Flow cytometry for microbead quantification

To validate the use of FC towards MB quantification, suspensions of MBs in both water and HA were analyzed against negative controls (blanks). The analysis of 1 mg L⁻¹ MBs in milliQ water is shown in Fig. 2A while the analysis of 1 mg L⁻¹ MBs in 10 mg L⁻¹ HA solution is shown in Fig. 2B. Each point on Fig. 2 represents an “event” detected by the instrument, categorized by its recorded fluorescent intensity (i.e. FITC-A) and its forward-scattering light intensity (i.e. FSC-A). The signal from the MBs occurs at elevated fluorescent intensities (~10⁴ to 10⁵ a. u.) regardless of the solution matrix chemistry. When just milliQ water is sampled, a small number of “contamination events” are noted at low fluorescent intensities (e.g. <10² a.u.). These can be considered as background noise and do not affect the detectability of the MBs which demonstrate much higher fluorescent intensities. When the solution matrix is switched to HA—more representative of complex solution chemistries seen in wastewater effluents—there are substantially more contamination events found at the low fluorescent intensities. These events may stem from the presence of HA aggregates which are detected by the instrument. Again, the detectability of the MBs remains unaffected.

To decisively categorize events as MBs, a four-sided gate was overlaid on the FSC-A vs. FITC-A plots. Its vertices are tabulated in Table S1 of the Supporting Information. Events within this gate, corresponding to the polygon formed to the right diagonal lines in Fig. 2, are assumed to be MBs. In Fig. 2A, 8,489 events were recorded during the analysis of the sample containing 1 mg L⁻¹ MBs in milliQ water, of which, 8,119 (95.6%) were assigned as MBs. With the milliQ water blank, 101 events were recorded, none of which (0%) were assigned as MBs. In Fig. 2B, 14,654 events were recorded in the analysis of the sample containing 1 mg L⁻¹ MBs in 10 mg L⁻¹ HA, of which, 8,842 (60.3%) were assigned as MBs. With the HA blank, 4,458 events were recorded and none of which (0%) were assigned as MBs.

Fig. 2 also shows the FITC-A and FSC-A adjunct histograms for each of the four data series which are plotted. That is, the number of events at each FITC-A intensity or FSC-A intensity are shown above/to the right of the plots, respectively. These histograms display the relative frequency distribution of FITC-A intensities (fluorescent “brightness”) and FSC-A intensities (akin to particle “size”) in each sample. For both the samples containing MBs, the FITC-A distributions show a median intensity value of approximately 10⁵ a.u., while in the MB-free samples, the

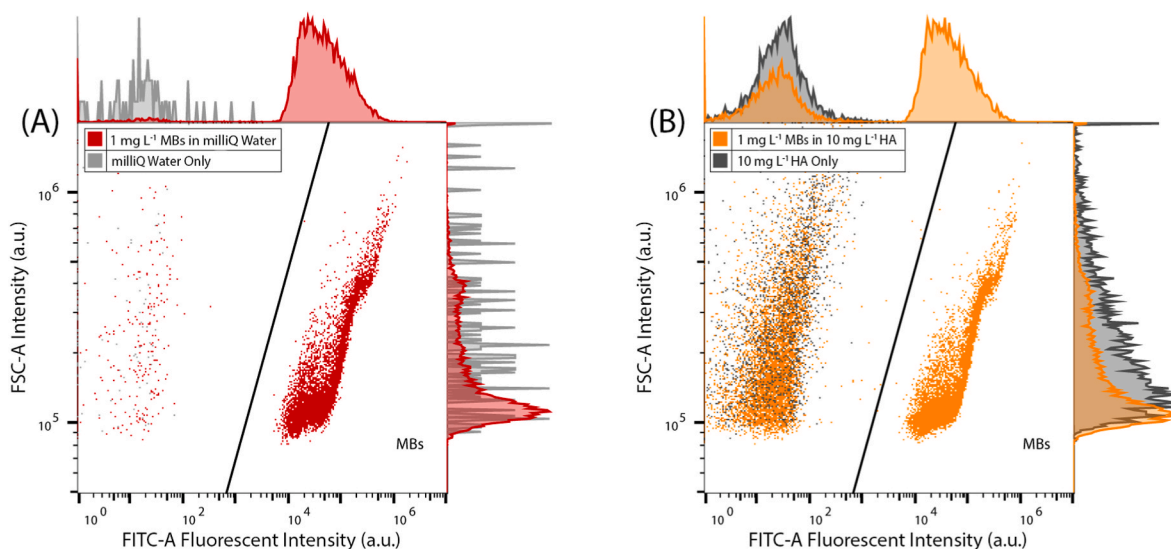


Fig. 2. Scatterplots and corresponding adjunct histograms for FSC-A versus FITC-A data collected using the FC instrument. (A) 1 mg L⁻¹ MBs suspended in milliQ water (■) and milliQ water blank (■). (B) 1 mg L⁻¹ MBs suspended in 10 mg L⁻¹ HA solution (■) and 10 mg L⁻¹ HA solution blank (■). Units for the axes are non-dimensional and a function of the instrument gains used (see Table S1). Counts in the polygon to the right of the diagonal line are gated as MBs. Histograms above the x-axes summarize the relative number of counts recorded at each FITC-A intensity. Histograms to the right of the y-axes summarize the relative number of counts recorded at each FSC-A intensity.

median fluorescent intensity ($\sim 10^1$ a.u.) is much lower in magnitude. These adjunct histograms confirm that MB events occur at much higher fluorescent intensities than the noise/contamination events. In the presence of MBs or HA, the FSC-A intensity distributions peak at relatively low values ($\sim 10^5$ a.u.), but have long, positive tails. The MBs and the contaminating events jointly contribute to a wide distribution of FSC intensities. But through sorting events by two parameters (*i.e.* FITC-A and FSC-A), it is trivial to resolve MB events from non-MB events in either solution chemistry. Thus, dividing the number of events gated as MBs by the known sample volume provides a facile measurement of the MB concentration.

In Fig. 2, it is recognized that no apparent shift in the FITC-A distributions can be seen when comparing MB samples suspended in either solution condition. A shift towards lower FITC-A values (*i.e.* lower fluorescent intensities) could imply substantial adsorption of HA onto the surfaces of the MBs (Alimi et al., 2018), attenuating fluorescence

emissions and affecting the gating/particle enumeration process. In this study, it is the fluorescent nature of the MBs that facilitates simple particle detection. Particles which are not inherently fluorescent may be dyed with a fluorophore (*e.g.* Nile Red) then analyzed in a typical fashion (Caputo et al., 2021; Kaile et al., 2020), however, the use of inherently fluorescent particles with a strong signature is expected to improve particle detection/resolution. Inherently fluorescent particles obviate the need for additional protocols such as particle dyeing and subsequent separation steps, as performed by Kaile et al. (2020).

3.2. A comparison of microbead quantification methods

To determine the ability of the FC instrument to resolve concentrations of MBs, calibration curves were constructed using MBs suspended in milliQ water or 10 mg L⁻¹ HA. MB suspensions were serially diluted from concentrations of 1 mg L⁻¹ down to 10⁻⁵ mg L⁻¹. Aliquots were

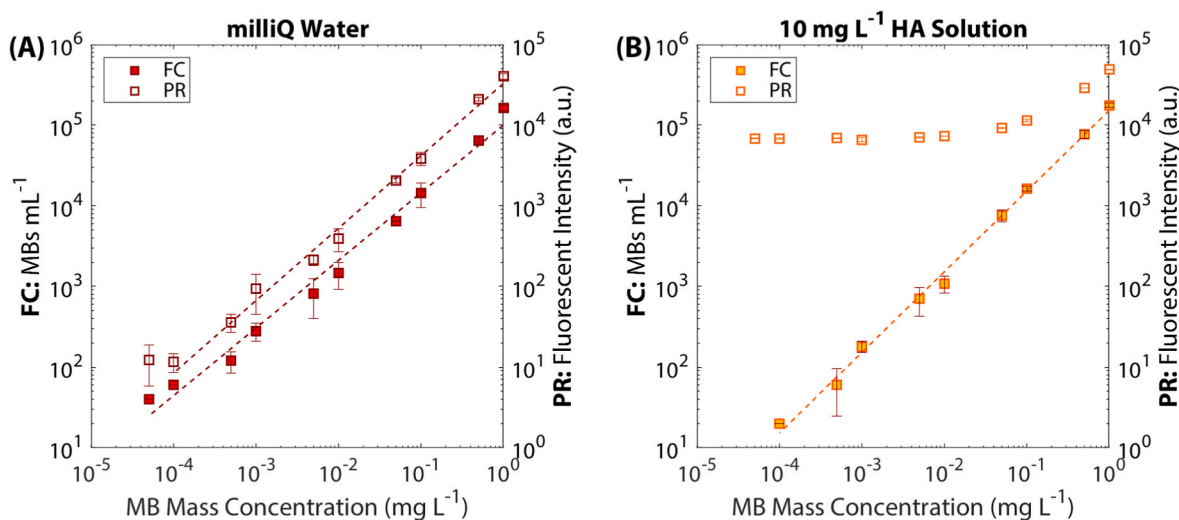


Fig. 3. Comparison of the solution composition when quantifying MB concentrations using the flow cytometer (FC; ■) and plate reader (PR; □). Conditions: (A) milliQ water and (B) aqueous 10 mg L⁻¹ HA solution. Power-law lines of best fit are shown (---), where appropriate. Error bars refer to one standard deviation about the average of triplicate measurements.

subsequently analyzed via FC and a “standard” fluorometric plate reader as a simple comparison. As shown in Fig. 3, plate reader results are reported in arbitrary units of fluorescent intensity, while FC results are reported in units of MBs mL⁻¹: the number of gated “MB events” per sample volume.

When milliQ water is used to suspend the MBs (Fig. 3A), the FC and plate reader obtain similar results. On the log-log plot, the relationship between the mass of MBs added to the solution matrix and the concentration via FC or plate reader fluorescent intensity is relatively linear and the relationships between the abscissa and ordinate variables are clear. At the lowest concentration depicted, individual MBs in a sample are counted (i.e. ~1 MB/50 μ L), obviating the need to extend the calibration curve to lower concentrations. At these very low concentrations, the data obtained by the plate reader deviates from the trend whereas the FC data maintains its linearity. Note that the standard deviations about the averages of triplicate measurements (i.e. denoted by the error bars) also tend to increase at lower MB concentrations (e.g. <10⁻³ mg L⁻¹), likely reflecting inhomogeneity of the particle suspensions. Overall, the FC instrument has at least as good of a resolution as the plate reader when the MBs are suspended in a pristine solution. But instead of providing a “bulk” correlation from fluorescent intensity to the mass concentration of MBs, the FC instrument records individual MB events such that number concentrations can be directly ascertained. In comparison to the standard “visual” MP quantification methods (e.g. microscopy), FC is much more efficient, with the analysis of a sample requiring only a few minutes to perform.

When the water matrix used to suspend the MBs is the 10 mg L⁻¹ HA solution (Fig. 3B), the performance of the FC instrument versus the plate reader is markedly different. The former instrument performs as before: MB concentrations are reliably resolved at concentrations lower than ~10⁻⁴ mg L⁻¹. However, the plate reader cannot resolve such concentrations under the current experimental design and the conditions that were tested (e.g. MB fluorophore used, concentration of HA, etc.). The trend plateaus at a fluorescent intensity of ~10⁴ a.u., indicating that MB concentrations are not resolvable below about 10⁻¹ mg L⁻¹: three orders of magnitude poorer than with the FC instrument.

This plateau in the plate reader fluorescent intensity versus MB mass concentration data was investigated to analyze the plate reader’s poor performance when HA was present. Using a second plate reader featuring a monochromator, 3D emission-excitation matrices (EEMs) were recorded for 10 mg L⁻¹ HA solutions spiked with decreasing concentrations of MBs, starting at 1 mg L⁻¹. Details regarding the procedure can be found in the Supporting Information. The results are shown in Fig. 4; note that concentration points in this figure were chosen to be a subset of the concentration points from Fig. 3. Over the range of excitation/emission wavelengths studied, the fluorescent intensities range from 10⁰ a.u. (blue) to $\leq 10^6$ a.u. (red). Note that where $\lambda_{ex} \approx \lambda_{em}$, very

high fluorescent intensities are recorded due to Rayleigh and Tyndall scattering (Murphy et al., 2013) and to the right of these “Rayleigh scattering lines”, data is not reported as the fluorescent intensities are meaningless where $\lambda_{ex} < \lambda_{em}$.

At the highest concentration of MBs (right panel), there is a broad, strongly fluorescent region with a maximum located at $\lambda_{ex} = 465$ nm and $\lambda_{em} = 500$ nm (3.5×10^4 a.u.) which corresponds to the emission of the MB fluorophore. The HA solutions also show moderate fluorescence (10^2 – 10^4 a.u.) at excitation/emission wavelengths away from the strongly fluorescent region over most of the spectrum that was analyzed. As the MB concentration decreases (i.e. right to left in the figure), the strongly fluorescent region corresponding to the MBs similarly decreases in intensity. Even in the absence of added MBs (far left panel), the HA solution produces these moderate fluorescent intensities across most of the spectrum of excitation/emission wavelengths that were studied, with higher fluorescent intensities seen towards the lower wavelengths. The fluorescent intensities of the HA solution alone exceed 10³ a.u. in the region where the MBs fluoresce (e.g. 2,182 a.u. at $\lambda_{ex} = 465$ nm/ $\lambda_{em} = 500$ nm) which explains the existence of the plateau in fluorescent intensity signal from the plate reader in Fig. 3B. At MB concentrations equal to, or below 0.05 mg MB L⁻¹, the EEMs are indistinct from the EEMs of solutions with no MBs added. A manifestation of this fluorescent behavior of HA has already been seen in Fig. 2B where a substantial number of events at low fluorescent intensities (e.g. ~10¹ a.u.) are recorded, attributable to the presence of HA aggregates.

Previous research has found that various HAs and humic substances exhibit fluorescent behavior over a wide range of wavelengths, as seen when the plate reader was used. Consequently, fluorescence spectroscopy is regularly used to characterize water and wastewater for its humic content (Carstea et al., 2016; Yang et al., 2015; De Melo et al., 2016). The chemical makeup and structure of HA and humic substances are highly complex and differ depending on the source; these molecules can contain a wide variety of chemical moieties including oxygen-bearing groups (e.g. alcohols, phenols, acids, quinones, ethers, etc.), nitrogenous heterocyclic rings, aliphatic chains, aromatic rings, or even sugars and peptides (De Melo et al., 2016; Schulten and Schnitzer, 1993). Some of these moieties can create regions of electron delocalization which can lead to the broadly fluorescent nature of the HA seen in these experiments. However, the HA fluorescent “fingerprint” is highly dependent on the HA source (Malcolm and MacCarthy, 1986), therefore fluorescence results from the HA used in this study may not coincide with fluorescence results from HAs originating from different sources. Furthermore, as HAs are only one common component of wastewater (Yang et al., 2015), the fluorescent fingerprint of the HAs may not be representative of wastewater as a whole nor should it be expected that all wastewaters would have the same fluorescent properties.

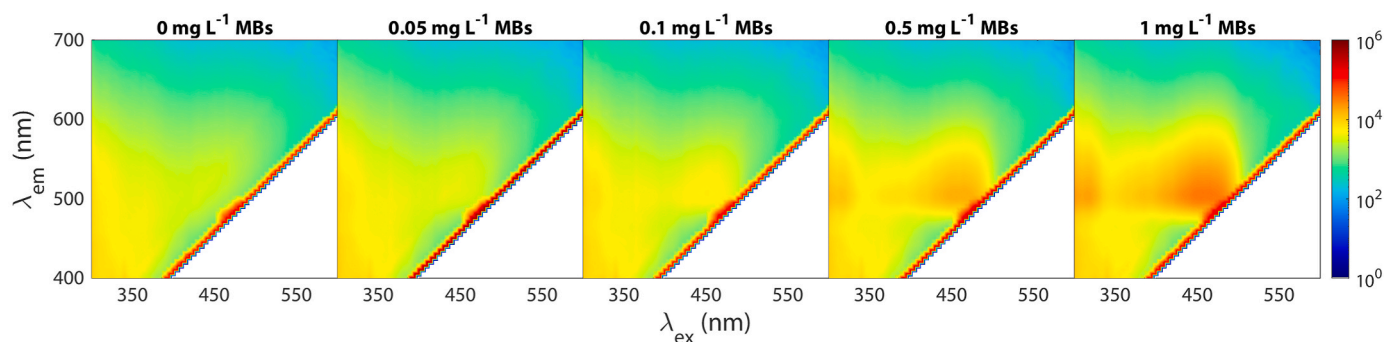


Fig. 4. Excitation-emission matrices illustrating the obscuring effect of HA when attempting to quantify MB concentrations using the plate reader. Magnitudes of fluorescent intensities are indicated by the colored legend, presented in arbitrary units. Suspensions of fluorescent MBs were diluted with HA solution, with MB concentrations decreasing from right to left. In all panels, the concentration of HA is 10 mg L⁻¹. White-coloured regions correspond to where $\lambda_{ex} < \lambda_{em}$. Panels showing additional MB concentrations are available in Figure S2 of the Supporting Information. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Evidently, the autofluorescent behavior of the HA presents a challenge when attempting to quantify the presence of aqueous fluorescent MPs using the plate reader, especially towards lower particle concentrations. This challenge is not encountered by the FC instrument as it uses measurements of both emitted fluorescence and FSC intensity (*i.e.* a two-parameter discrimination) to differentiate between particles which both fluoresce and scatter light, and the background fluorescence from the HA. This demonstrates the potential of the technique towards measuring MP concentrations in complex solution matrices. To this end, Wang et al. (2021) briefly confirm that fluorescent MP concentration measurements can also be made successfully via FC in river water and WWTP effluent solution matrices, though they do not appear to follow up on these findings in the context of MP removal processes.

3.3. Microbead removal via membrane filtration

Our results indicate that FC is amenable towards quantifying fluorescent MB concentrations, even in the presence of HA. We have opted to demonstrate the utility of the technique in MP removal experiments involving MF membranes. The 1–5 μm MBs were filtered through membranes with 0.45 and 5 μm nominal pore sizes. Filtration performance was compared between the cases where milliQ water and 10 mg L^{-1} HA solutions were used as the solution matrices. Results from the hour-long filtration experiments are summarized in Fig. 5 using FC to measure MB concentrations in feed and permeate samples. Two replicates are shown at each condition; it should be noted the results from both replicates are consistent with each other.

3.3.1. Filtration experiments in milliQ water

When the 0.45 μm pore size membrane and milliQ water matrix is

considered, very little MB transmission is noted, as evidenced in Fig. 5A. Concentrations of MBs in permeate samples were $\leq 2 \times 10^2 \text{ MB mL}^{-1}$, originating from feeds with concentrations of $(1.30\text{--}1.36) \times 10^5 \text{ MB mL}^{-1}$. Furthermore, the transmission is notably sporadic: only 39% and 52% of permeate samples in the two trials, respectively, contained any MBs. This is expected as the nominal pore size is smaller than the MB particles. In contrast, when the 5 μm pore size membrane and milliQ water matrix is considered, there is a greater transmission of MBs (Fig. 5B): permeate concentrations were found to be $< 1.05 \times 10^5 \text{ MB mL}^{-1}$, originating from feeds with concentrations of $(1.37\text{--}1.67) \times 10^5 \text{ MB mL}^{-1}$. Accordingly, maximum permeate concentrations from the 5 μm membranes are approximately 700 times higher than the maximum permeate concentrations from the 0.45 μm membranes. This is expected as the nominal membrane pore size is on the order of magnitude of the MBs' sizes, permitting particle transmission, assuming that MB rejection is due to purely size-exclusion means. Over the course of these filtrations with the 5 μm membrane (Fig. 5B), declining concentrations of MBs in the permeate samples point to fouling of the membrane's surface/pores by the MBs, which is inevitable due to the dead-end filtration mode that was used.

3.3.2. Filtration experiments in humic acid solution

When the experiments from §3.3.1 were repeated in the presence of HA, MB transmission was reduced regardless of the membrane that was used. For the 0.45 μm pore size, the maximum MB concentration fell to $4 \times 10^1 \text{ MB mL}^{-1}$ in the permeate samples (Fig. 5D). It should be noted that such "transmission" may even manifest as noise or sample contamination within the FC instrument. This maximum permeate concentration is equivalent to 2 MBs per 50 μL sample which is within the expected intra-sample variability at these concentrations (standard

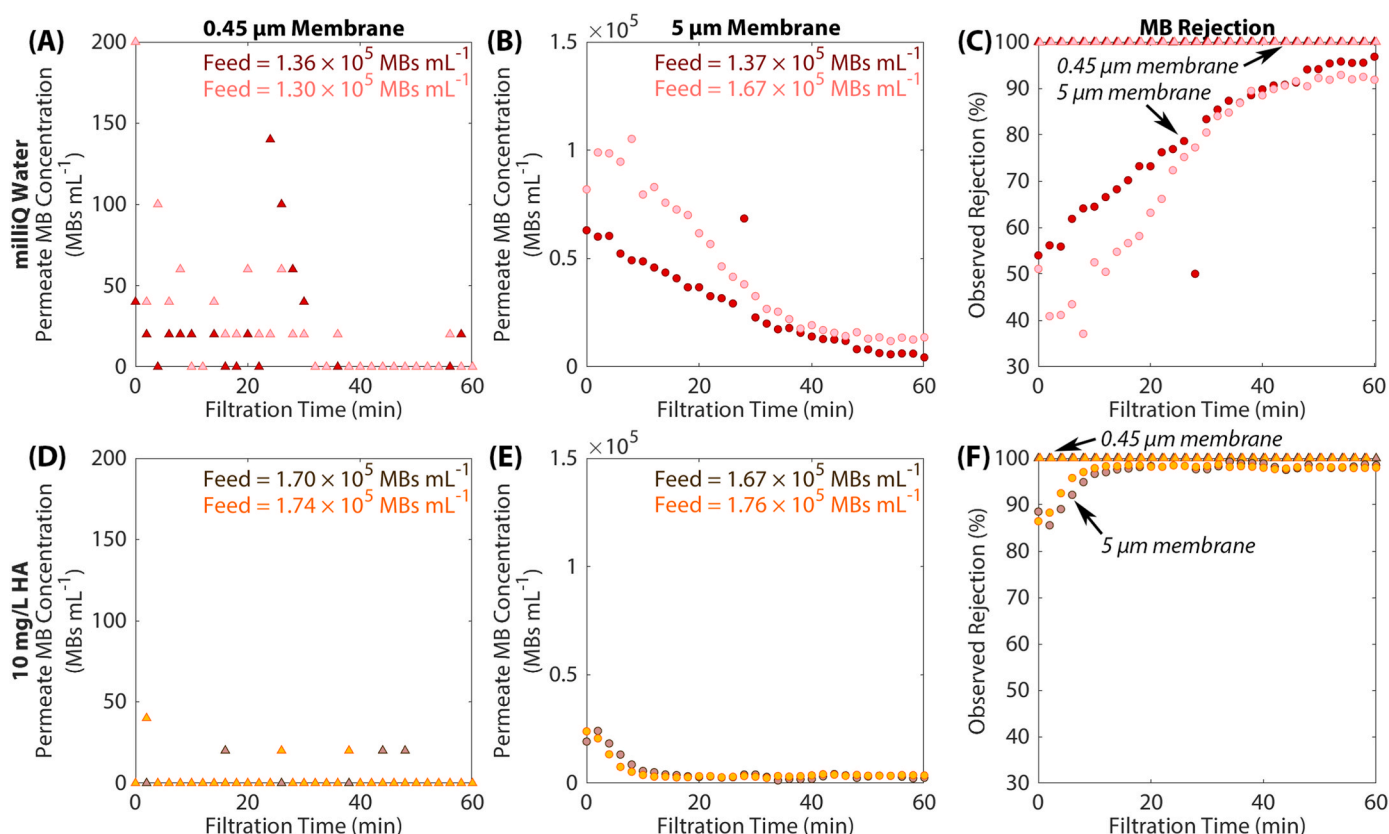


Fig. 5. Comparison of the effect of pore size on MB transmission and observed rejection using the Durapore® 0.45 μm (▲) and 5 μm (●) membranes operated at a constant permeate flux of 1,000 LMH. The top and bottom rows, respectively, summarize experiments in milliQ water and 10 mg L^{-1} HA solutions. MB counts were obtained via FC to produce MB concentrations/rejection coefficients. Experiments with each of the membranes were performed in duplicate; individual colours in the figure indicate the replicates. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

deviation = 2 MBs per sample). As such, the transmission seen is indistinguishable from background noise. Analysis of larger permeate samples would help elucidate these effects on the instrument's MB detection limits. Furthermore, transmission was even more sporadic with only around 10% of samples collected containing measurable quantities of MBs.

Considering the 5 μm membrane, MB transmission decreased substantially in the presence of HA (Fig. 5E) versus in its absence (Fig. 5B). The MBs concentrations in permeate samples did not exceed 2.5×10^4 MB mL⁻¹ which is approximately four times less than when the milliQ water matrix was used. The MB concentration in the last sample that was taken (*i.e.* $t = 60$ min) in the presence of HA was $<3.54 \times 10^3$ MB mL⁻¹ versus $<1.36 \times 10^4$ MB mL⁻¹ when milliQ water was used. The strong overall reduction in permeate MB concentrations in the presence of HA can be attributed to membrane fouling caused by the HA which adsorbs to the membrane surface or inside the pores, restricting the ability of MBs to pass through the membrane. Fouling of MF membranes by HAs in this manner is well-documented in the literature (Yuan and Zydney, 2000), with similar phenomena seen when wastewater is filtered. Now whereas the decline in MB concentrations in permeate samples over time is attributable to MB-induced fouling when milliQ water was used (Fig. 5B), this decline can be attributed to both MB- and HA-induced membrane fouling when HA was present (Fig. 5D). In the latter case, HA adsorption onto both membrane/MB surfaces is likely (Alimi et al., 2018), but recent research by Pradel et al. (2021) suggests that in the presence of HA, MP aggregation can be reduced as adsorbed HA contributes to electrostatic repulsion, while larger HA molecules can sterically hinder aggregation.

From Fig. 5D, it is also interesting to note that the MB concentrations in the 5 μm membrane permeate samples reached a non-zero plateau in the presence of HA, suggesting that the fouling layer does not completely occlude the membrane pores but constricts them enough to reduce MB transmission. When comparing the results of the filtrations in milliQ water versus HA solution, solution chemistry presents as an important factor which drives the effectiveness of the removal of the MBs, thus MP filtration experiments should be conducted in reasonable solution conditions for the results to be representative of real-world processes. In WWTPs, fouling by organic matter and/or MPs may form a dynamic membrane which could aid in preventing the transmission of MPs into the permeate stream.

3.3.3. A comparison of rejection coefficients

Data from the experiments shown in Fig. 5 was subsequently used to calculate the number-based MB observed rejection coefficients R_n for each of the permeate concentration values shown in Fig. 5, where n_f and n_p are the number concentrations of MBs in the membrane's feed and permeate streams, respectively:

$$R_n = \left[1 - \frac{n_p}{n_f} \right] \times 100\% \quad (3)$$

Note that n_f refers to the initial concentration of the suspension fed to the membrane cell, with no corrections made for concentration polarization in the dead-end filtration.

The data in the "MB Rejection" panels of Fig. 5 are directly translated from the neighboring "0.45 μm Membrane" and "5 μm Membrane" panels. When the milliQ water matrix is considered, R_n values for the 0.45 μm membrane exceeded 99.2%. The MBs are strongly rejected by the smaller membrane pores, however, the rejection is not complete. For the 5 μm membrane, R_n values ranged from 40–60% at the start of the filtration experiments, but steadily climbed beyond 90% towards the 1 h mark as particle fouling restricted the transmission of the MBs. It is interesting to note that the MB manufacturer states that 90% of the particles fall less than 5 μm in size (*i.e.* the d_{90} value), however, transmission of 90% of the MBs is not seen by the 5 μm membrane, therefore this rating does not adequately reflect the size of these MBs that can be

rejected.

The effects of HA fouling further increase the rejection coefficients, with R_n exceeding 99.5% for the 0.45 μm membrane, versus 85–90% initially for the 5 μm membrane, rapidly climbing to >95% by the sample taken at the 10 min mark. The implication is that performing filtration experiments in pristine solution matrices (*e.g.* water, buffer, etc.) is not sufficient to elucidate all filtration phenomena. Moreover, if results are sought which should mimic filtration performance in real WWTPs, there is an inherent need to also use solution matrices that mimic those in WWTPs.

The results from the MP removal experiments in this work demonstrate the inherent effectiveness of membranes for MP removal in comparison to other technologies. Using the previously described work by Wang et al. (2021) as an example, with exception to their ferrate oxidation process, modest microsphere removal rates were typically seen: approximately 30–70% for media filtration and approximately <5–90% for coagulation processes. Pulido-Reyes et al. (2022) noted similar coagulation/filtration findings in the removal of Pd-labelled polystyrene NP particles (~215 nm), however, up to a 99.9% removal was found to be possible when consecutive rapid sand filtration, activated carbon filtration, and slow sand filtration processes were employed. In comparison with the results from this study, a single 0.45 μm membrane demonstrated similar MP removal performance, where almost complete MP removal (>99%) was seen regardless of the solution conditions that were used. Even the 5 μm membrane provided similar MP removal rates in the presence of HA. Both efficient and effective MP removal performance for particles ~1 μm and larger could feasibly be achieved using a 0.45 μm pore size MF membrane in tertiary wastewater filtration processes.

3.3.4. Microbead size estimates in filtration experiments

It is noted that MP size influences the ecological consequences of these particles, where smaller particles may pose a greater risk than larger particles based on their elevated specific surface area and propensity for ingestion (Enfrin et al., 2021; Chae and An, 2017). There is substantial value in obtaining MP size measurements in conjunction with MP particle concentration measurements in order to track the fate of particles of different sizes in MP removal studies. In §2.5, we adapted a model described by Julià et al. (2000) towards estimating the size of particles using FSC-A measurements provided by a flow cytometer. Using the process illustrated in Figure S3, a function $d = \varphi(i)$ was created to map FSC-A intensities values obtained via FC to corresponding MB particle diameters obtained *ex situ* via FESEM. The model data was truncated just before $i = 4 \times 10^6$ a.u. to avoid creating a non-smooth $\varphi(i)$ at high i due to the very small number of particles with very high intensities (*i.e.* outliers). The final model is thus valid for $i \in [7 \times 10^4, 4 \times 10^6)$ and is shown in Figure S4 of the Supporting Information. Using this model to obtain particle size estimates in a sample from a filtration experiment, the FSC-A intensities corresponding to the gated events were recorded. Each intensity i^* was passed to the model function $d = \varphi(i)$ which produced an estimate for the particle size d^* from that event. As the model function was produced using CDFs formed from discrete data, linear interpolation was used between points in the model.

From this model, size estimates were obtained of the MBs in feed/permeate samples from the filtration experiments shown in Fig. 5. First to verify the model's suitability, particle size distributions for the MBs measured *ex situ* were overlaid with particle size distributions for the MBs in feed samples estimated via the FC model. The overlays, shown in Figure S5 of the Supporting Information, demonstrate that the distributions produced by both of these sizing methods are in good agreement. Therefore, the model is valid for use in the systems described in this work. The data in Figure S5 also shows no indication of substantial changes in particle size when HA was present versus when it was absent. Evidence of MB aggregation or a thick adsorbed HA layer on the

particles would reasonably manifest as a rightward shift in the estimated particle size distributions in the presence of HA. No appreciable shift is noted. The particle size distributions are in agreement, regardless of the solution conditions that are used.

Subsequently, Fig. 6 focuses on the estimated particle size distributions for MBs in the feed and permeate samples where the 5 μm membrane and 10 mg L^{-1} HA solutions were used. Note that only Replicate #1 is shown and discussed, here, as the variability between replicates is low (see Fig. 5). Results for Replicate #2 can be found in Figure S6 and the subsequent tables in the Supporting Information. In the feed sample, 8366 particles were analyzed via FC and the particle size distribution of these MBs has a positive skew: a greater number of events are seen corresponding to smaller particle sizes. In Replicate #1, the estimated median particle size ($\tilde{d}$) is calculated to be 1.24 μm , with estimated d_{10} and d_{90} sizes found to be 0.61 μm and 2.24 μm , respectively. For the purposes of brevity, permeate samples were analyzed every 20 min for the duration of the experiment. The particle size distributions shift noticeably towards smaller sizes in the permeate versus feed samples. For example, while the feed sample had a median particle size of 1.24 μm , the equivalent value in the permeate samples decreased to 0.82 μm (averaged over the four analyzed samples): a 33% decline, on average. Similar findings can be seen when the d_{10} and d_{90} values of the distributions are considered, as summarized in Table S2 of the Supporting Information.

While the decline in size metrics proceeding from the feed to the permeate samples is also evident when milliQ water suspended the MBs, this preferential transmission of smaller particles sizes is most pronounced in the presence of HA, implying HA's role in the fouling-mediated rejection of larger MBs. To summarize these effects, MB size data from all the filtration experiments using the 5 μm membrane, including both solution conditions, is presented in Table S3 and Figure S7 of the Supporting Information.

For particle sizes near to (or smaller than) the illuminating wavelength, Mie scattering theory predicts that the intensity of light scattered at small angles increases with the sixth power of a particle's size, though empirical experiments find this exponent to be smaller than its theoretical prediction (Chandler et al., 2011; Koch et al., 1996). The expectation created is that FSC intensity should increase non-linearly with increasing MB size. For the system described in this work, a graph of FSC-A versus estimated particle size indeed produces a non-linear, monotonically increasing relationship with a concave-upwards shape. This fits in accordance with theory and with results obtained by Foladori et al. (2008) who developed a correlation between FSC intensity measurements and the size of silica beads. In contrast, Julià et al. (2000) found that the FSC intensity versus particle diameter function was concave-downwards when actual bacteria were sized. These bacteria differ in refractive properties, size, and geometry from the spherical polymer MBs used in this study. Any discrepancies, here, likely arise from the fact that the exact FSC-A values produced are

a complex function of particle properties, the sensitivity/linearity of the FC detectors, and the calibration/alignment of the instrument optics (Foladori et al., 2008; Chandler et al., 2011).

Notwithstanding, the noted variations in concavity are likely to be immaterial to the results presented in this work. It should be cautioned that even if a (non-linear) relationship between the scattering intensity and particle size can be generated, such a relationship is not universal and is a function of system parameters (Galbusera et al., 2020). This implies that new calibrations would need to be generated for different instruments, different MP shapes/sizes/compositions, and different solution chemistries. Where the MPs have diverse refractive indices or optical properties (e.g. absorption of light by pigments), the solutions contain various solids, or there is a propensity for solid matter (e.g. NOM, biofilms) to deposit on the MPs, it would be expected that different relationships between FSC-A intensity and MB size would exist. These relationships could even change over time, such as due to the transient growth of biofilms. The technique is thus limited in that the properties of the MPs and solution matrix should be controlled for in advance.

Whereas the model presented here was specifically formulated for estimating the size of the “regular” MPs used in this study, a foreseeable challenge arises when attempting to estimate the size of MPs originating from heterogeneous populations, especially those that are irregular in shape or have variable surface properties. Such variations would be reasonably expected to affect light scattering by these particles, and FSC intensity measurements may not as readily correlate to MP size. However, SSC intensity measurements relate beyond particle size (van Gaal et al., 2010; Tzur et al., 2011) to granularity/complexity and may prove useful in accounting for irregularity arising from the presence of oblique surfaces or surface roughness. Taking an example from biological systems, yeast and bacterial cells can be distinguished by their FSC and SSC intensity signals which differ based on cell size and shape (Díaz et al., 2010; Veal et al., 2000). Employing SSC intensity measurements concurrently with FSC and fluorescent intensity measurements (i.e. a three-parameter discrimination) may be useful for quantifying the presence of irregular-shaped particles and formulating new size estimation models, as well as for performing investigations in complex solution matrices.

Alternative methodologies may also be employed in these circumstances. Instead of using light scattering intensity to estimate the sizes of MPs, Hyeon et al. (2024) used a flow cytometer with a FlowCam device to obtain images which were subsequently analyzed to obtain geometric and morphological properties of the fibers (e.g. “size”) in filtration samples. Of substantial value, the researchers quantified how fibers which had a longer geodesic length, lower straightness, and demonstrated a greater curl were more highly rejected due to more effective size exclusion. This technique may present as a viable alternative for some systems.

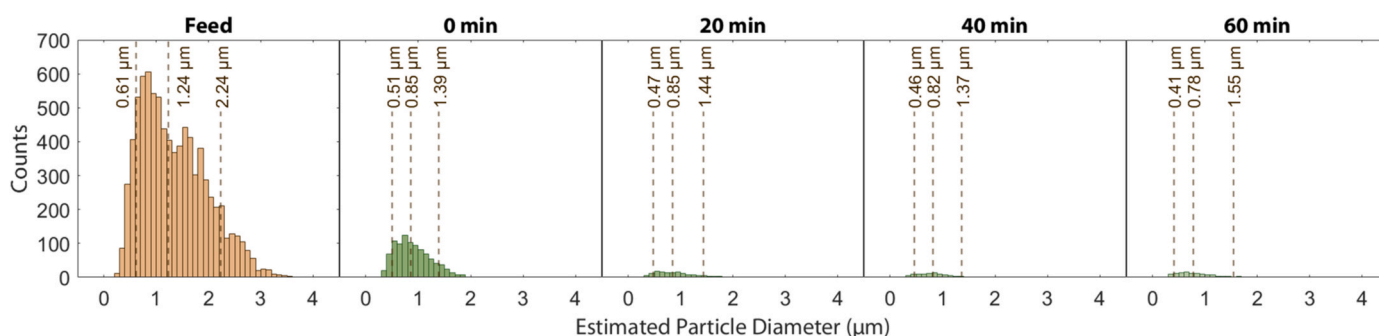


Fig. 6. Estimation of MB particle sizes (μm) in selected permeate samples for the filtration of 1 mg L^{-1} MBs in 10 mg L^{-1} HA solution using the 5 μm membrane. Only Replicate #1 is shown. Estimated particle sizes from feed samples (■) versus permeate samples (■) taken at 20-min intervals are shown. From left to right, the d_{10} , median, and d_{90} statistics of the particle size distributions are included (—).

4. Conclusions

Measuring the concentrations of MP particles presents a roadblock in the acceleration of MP-related research. There is a clear opportunity to develop and validate techniques to more efficiently measure the presence of MPs and their properties, especially in dilute samples containing very small particles. This work sought to apply FC—an analytical technique heavily used outside of the field of MPs—to efficiently characterize fluorescent spherical plastic MBs in aqueous samples. Even in the presence of HA, FC discerned and quantified concentrations of the MBs down to $\sim 10^{-4}$ mg L⁻¹. When a plate reader was used in comparison, the autofluorescent nature of the HA across a broad range of excitation/emission wavelengths overwhelmed the signal associated with the MB fluorophore at concentrations below ~ 0.05 mg L⁻¹. It is possible that the effective operating window of the plate reader may widen if a different MB fluorophore is chosen whose emission wavelength corresponds to very low HA fluorescent intensities (*i.e.* long wavelengths). Regardless, FC demonstrated to be an effective MB quantification tool, even at low MB concentrations and in the presence of autofluorescent background matter.

Furthermore, the individual events recorded via FC correspond to the detection of individual particles in samples and are not “bulk” measurements of fluorescent matter in the sample, as is the case with the plate reader. Recording light scattering intensity data from individual particles allowed a simple model to be constructed which correlates FSC-A intensity values to MB size. This model was leveraged in subsequent MB filtration experiments whereby FC was exploited to measure MB concentrations and properties in both membrane permeate and feed samples. Very high MB retention was seen with the 0.45 μ m membrane, whereas MB rejection proved to be transient for a 5 μ m membrane, depending on the solution conditions and filtration time. The presence of HA was found to mediate particle transmission, implying that this NOM can enhance particle rejection, though at the expense of membrane fouling. During filtrations using the latter membrane, significant reductions in median particle size were seen, implying a preferential rejection of larger MBs and a transmission of smaller particles. This is particularly concerning from an environmental standpoint as it is expected that smaller particles pose a greater ecological hazard. To our knowledge, this study represents the first time that FC has been used to estimate the size of “environmental” MP particles as a part of a membrane MF process. Similarly to how other researchers have utilized FC to measure cell biovolume, this technique could be extended in a future work to compute MP mass from particle volume estimates, or modifications to the methodology may be pursued when dealing with heterogeneous populations of MPs or irregularly-shaped particles.

In this work, the utility of the FC technique was demonstrated specifically for quantifying the presence of well-defined MP particles suspended in one of two solution conditions as part of a MF process. Given that FC is already deployed in the biomedical and environmental sciences to count cells of various of shapes and sizes, it is feasible that the technique could be extended to systems involving regular/irregularly-shaped fluorescent MPs of various shapes and sizes—provided that the particles fall within the operating window of the instrument. The technique may be feasibly leveraged in systems with complex solution chemistries (*e.g.* municipal wastewater) and in the evaluation of other MP removal technologies. Ongoing research should be undertaken to rigorously validate FC in these applications.

CRediT authorship contribution statement

Ryan J. LaRue: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Samuel Koo:** Methodology, Investigation. **Ashleigh Warren:** Visualization, Methodology, Investigation. **Yves G. McKay:** Visualization, Investigation. **David R. Latulippe:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ryan LaRue reports financial support was provided by Vanier Canada Graduate Scholarships and by Ontario Graduate Fellowships. Ashleigh Warren reports financial support was provided by Natural Sciences and Engineering Research Council of Canada and by ECO Canada. David Latulippe reports financial support was provided by Ontario Ministry of Research, Innovation and Science. Ryan LaRue reports membranes were provided by EMD Millipore.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2024.143613>.

Data availability

Data will be made available on request.

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